

Aref Pariz, Ph.D.

PROFESSIONAL PROFILE

Computational and theoretical neuroscientist with training in physics and postdoctoral experience in brain stimulation, neural dynamics, TMS-EEG, tACS, effective connectivity, and individualized modeling of brain networks. Current research focuses on computational psychiatry and precision non-invasive brain stimulation in clinical populations.

RESEARCH AREAS

Computational psychiatry and computational neuroscience
Non-invasive brain stimulation: TMS, theta burst stimulation, tACS, and tES/tCS
Neural dynamics, oscillations, information transfer, and effective connectivity
TMS-EEG, EEG analysis, fMRI-constrained modeling, and network neuroscience
Synaptic plasticity, neural heterogeneity, resilience, and brain-state dependence
Machine learning, data analysis, and individualized stimulation protocols

ACADEMIC APPOINTMENTS AND RESEARCH EXPERIENCE

Feb. 2026-present	Assistant Instructor, Department of Psychiatry, UT Southwestern Medical Center, Dallas, TX, USA Research focus: individualized tES/tACS and related stimulation methods for addictive and related conditions.
Sep. 2023-Jan. 2026	Postdoctoral Researcher, Institute of Mental Health Research at The Royal and University of Ottawa, Ottawa, ON, Canada Mentors: Dr. Sara Tremblay and Prof. Jeremie Lefebvre. Research focus: effects of TMS and theta burst stimulation on brain disorders; TMS-EEG analysis and effective connectivity.
Jan. 2021-Aug. 2023	Postdoctoral Researcher, Department of Biology, University of Ottawa, Ottawa, ON, Canada Mentor: Prof. Jeremie Lefebvre. Research focus: effects of brain stimulation on synaptic modifications and neuronal population dynamics.
Dec. 2019-Dec. 2020	Postdoctoral Researcher, Department of Chemistry, Institute for Advanced Studies in Basic Sciences, Zanjan, Iran Mentor: Prof. Abdollahi. Research focus: chemometric data analysis methods in neuroscience.
Apr. 2018-Mar. 2019	Visiting Researcher, IFISC (UIB-CSIC), Palma de Mallorca, Spain Mentor: Prof. Claudio R. Mirasso. Research focus: effects of communication delay on information transfer.
Sep. 2012-Mar. 2018	Ph.D. Researcher, Institute for Advanced Studies in Basic Sciences, Zanjan, Iran Mentor: Prof. Alireza Valizadeh.
Sep. 2009-Mar. 2012	M.Sc. Researcher, University of Zanjan, Zanjan, Iran Mentor: Dr. Saeed Ghanbari.

EDUCATION

2012-2018	Ph.D. in Physics, Institute for Advanced Studies in Basic Sciences, Zanjan, Iran Dissertation: Controlling the information transfer by phase manipulation of population activity of neuronal networks. Supervisor: Dr. Alireza Valizadeh.
2009-2011	M.Sc. in Physics, University of Zanjan, Zanjan, Iran Thesis: Collapse of Dipolar Bose-Einstein Condensation. Supervisor: Dr. Saeed Ghanbari.
2005-2009	B.Sc. in Physics, Shahid Bahonar University of Kerman, Kerman, Iran

FUNDING AND FELLOWSHIPS

2026	Brain & Behavior Research Foundation, NARSAD Young Investigator Grant Title: Optimized Frontoparietal Synchronization to Reduce Craving in Methamphetamine Use Disorder:
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2026	A Concurrent tACS-fMRI Trial. Role: Principal Investigator. Status: Under Review. NIH R01 Title: Optimized Frontoparietal Synchronization to Reduce Craving in Methamphetamine Use Disorder: A Concurrent tACS-fMRI Trial. Role: Key Personnel. Status: Under Review.
2019	Postdoctoral Fellowship Grant, Iranian Elite Federation
2018	Visiting Fellowship Grant, IFISC, University of the Balearic Islands, Palma de Mallorca, Spain
2017	Study Grant, Institute for Research in Fundamental Sciences, Tehran, Iran

PEER-REVIEWED PUBLICATIONS

11. Asl, K. J., Naseri, A., Amjadi, M., Alavi, S. M. S., **Pariz, A.**, Tatardar, F., & Rasoulzadeh, F. (2026). Nanozymatic platform for dual colorimetric and fluorimetric sensing of H₂O₂ and glucose using Ni-CoWO₄ nanoparticles. *Scientific Reports*. <https://doi.org/10.1038/s41598-026-58001-3>
10. Lefebvre, J., **Pariz, A.**, Thivierge, J.-P., & Hutt, A. (2025). Surfing brain waves without sinking: A commentary on "Entrainment by transcranial alternating current stimulation: Insight from models of cortical oscillations and dynamical systems theory" by M. Madadi Asl and A. Valizadeh. *Physics of Life Reviews*, 55, 55-57. <https://doi.org/10.1016/j.plprev.2025.08.009>
9. Trotter, D. J., **Pariz, A.**, Hutt, A., & Lefebvre, J. (2025). Morphological variability may limit single-cell specificity to electric field stimulation. *Frontiers in Synaptic Neuroscience*, 17, 1621352. <https://doi.org/10.3389/fnsyn.2025.1621352>
8. Lefebvre, J., & **Pariz, A.** (2025). Amplifying post-stimulation oscillatory dynamics by engaging synaptic plasticity with transcranial alternating current stimulation. *Frontiers in Network Physiology*, 5, 1621283. <https://doi.org/10.3389/fnetp.2025.1621283>
7. Hutt, A., Trotter, D., **Pariz, A.**, Valiante, T. A., & Lefebvre, J. (2024). Diversity-induced trivialization and resilience of neural dynamics. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 34(1), 013147. <https://doi.org/10.1063/5.0165773>
6. **Pariz, A.**, Trotter, D., Hutt, A., & Lefebvre, J. (2023). Selective control of synaptic plasticity in heterogeneous networks through transcranial alternating current stimulation (tACS). *PLOS Computational Biology*, 19(4), e1010736. <https://doi.org/10.1371/journal.pcbi.1010736>
5. Yousefi, N. N., Mortezapour, A., Naeimi, G., Nosrati, F., **Pariz, A.**, & Lo Franco, R. (2022). Quantum enhancement of qutrit dynamics through driving field and photonic-band-gap crystal. *Physical Review A*, 105(4), 042212. <https://doi.org/10.1103/PhysRevA.105.042212>
4. Sanchez-Claros, J., **Pariz, A.**, Valizadeh, A., Canals, S., & Mirasso, C. R. (2021). Information transmission in delay-coupled neuronal circuits in the presence of a relay population. *Frontiers in Systems Neuroscience*, 15, 705371. <https://doi.org/10.3389/fnsys.2021.705371>
3. **Pariz, A.**, Fischer, I., Valizadeh, A., & Mirasso, C. R. (2021). Transmission delays and frequency detuning can regulate information flow between brain regions. *PLOS Computational Biology*, 17(4), e1008129. <https://doi.org/10.1371/journal.pcbi.1008129>
2. **Pariz, A.**, Esfahani, Z. G., Parsi, S. S., Valizadeh, A., Canals, S., & Mirasso, C. R. (2018). High frequency neurons determine effective connectivity in neuronal networks. *NeuroImage*, 166, 349-359. <https://doi.org/10.1016/j.neuroimage.2017.11.014>
1. Mohammadi, A., Ghanbari, S., & **Pariz, A.** (2013). A two-dimensional permanent magnetic lattice for ultracold atoms. *Physica Scripta*, 88(1), 015601

SUBMITTED MANUSCRIPTS, PREPRINTS, AND MANUSCRIPTS IN PREPARATION

7. Hashemi, M., Shi, W., Soleimani, G., Villanueva-Miranda, I., **Pariz, A.**, Anteraper, S., Jin, J., Mansour L., S., Benette Marques, D., Mirjalili, M., Opitz, A., Juliano, A., Triebkorn, P., Wang, H., Yang, C., Rajji, T., Brunoni, A. R., Jirsa, V., Xiao, G., & Ekhtiari, H.. Artificial Intelligence for Precision Brain Stimulation: From Neuroimaging to Foundation Models and Virtual Brains [Preprint]. PsyArXiv preprint. https://doi.org/10.31234/osf.io/qnjd6_v1
6. **Pariz, A.**, Lefebvre, J., & Tremblay, S.. NESTApp: A GUI-Based MATLAB Application for Simplified TMS-EEG Preprocessing with EEGLAB [Preprint]. SSRN preprint. <https://ssrn.com/abstract=6204470>
5. Hutt, A., McCulloch, M., **Pariz, A.**, & Lefebvre, J.. Noise-like fluctuations drive shifts in 1/f-alpha power-law dynamics associated with changes in brain states [Manuscript]

4. Sharifzadeh, S., Jouzdani, A. F., Souki, A., Alekseichuk, I., Brunoni, A. R., Marques, D. B., Ebrahimi, M., Ghobadi-Azbari, P., **Pariz, A.**, Soleimani, G., Yavari, F., Khosrowabadi, R., Violante, I. R., & Ekhtiari, H.. Personalized Brain Stimulation Based on Brain Oscillations: A Systematic Review of 69 Magnetoencephalography- and Electroencephalography-Informed Transcranial Alternating Current Stimulation (tACS) Trials [*Preprint*]. PsyArXiv preprint. https://doi.org/10.31234/osf.io/q8x6v_v1
3. **Pariz, A.**, Hadas, H., Lepage, M., Daskalakis, Z. J., Lefebvre, J., & Tremblay, S.. Network-Level Responses to iTBS Pulse Dose: A Non-Linear Pattern Detected with TMS-EEG [*Manuscript*]
2. O'Sullivan, D., **Pariz, A.**, Talidou, A., Trotter, D., & Lefebvre, J.. Myelin-mediated control on neural excitability, correlations and synaptic plasticity [*Manuscript in preparation*]
1. **Pariz, A.**, Hutt, A., Valiante, T., Lankarany, M., Skinner, F., Griffiths, J., & Lefebvre, J.. Silence Through Coherence [*Manuscript in preparation*]

CONFERENCE PROCEEDINGS AND PUBLISHED ABSTRACTS

6. **Pariz, A.**, Lefebvre, J., & Valizadeh, A. (2021). Phase response curve of the excitatory-inhibitory neuronal populations and its role in the coherent oscillation of interconnected brain regions. *Journal of Computational Neuroscience*, 49(Suppl. 1), S188-S188
5. Sanchez-Claros, J., Mirasso, C. R., Kang, M., **Pariz, A.**, & Fischer, I. (2019). Information transmission in delay-coupled neuronal circuits. *CNS2019, BMC Neuroscience*, 20, P81
4. Mirasso, C. R., **Pariz, A.**, Canals, S., & Valizadeh, A. (2019). High frequency neurons help routing information in brain networks. *CNS2019, BMC Neuroscience*, 20, P320
3. **Pariz, A.**, Parsi, S. S., & Valizadeh, A. (2016). High frequency neuron can facilitate propagation of signal in neural networks. *CNS2016, BMC Neuroscience*, 17, P42
2. Mohammadi, A., Ghanbari, S., & **Pariz, A.** (2010). Direct forced evaporative cooling in permanent magnetic microtraps. arXiv:1007.2912. <https://arxiv.org/abs/1007.2912>
1. **Pariz, A.** (2011). The Student Guide to General Physics. Book in Persian

INVITED TALKS AND ORAL PRESENTATIONS

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| June 2025 | Invited Speaker, University of Minnesota, Minneapolis, MN, USA
Talk: Modulating brain circuits and neuronal population activity through tACS. |
| Oct. 2023 | Speaker, Canadian Computational Neuroscience Spotlight, Canada
Talk: Selective control of synaptic plasticity and stabilizing post-stimulation oscillatory dynamics in neuronal populations. |
| June 2023 | Invited Speaker, CAIMS Annual Conference, University of New Brunswick, NB, Canada
Talk: Selective control of synaptic plasticity and stabilizing post-stimulation oscillatory dynamics in neuronal populations. |
| June 2022 | Speaker, Canadian Computational Neuroscience Spotlight, Canada
Talk: Selective engagement of synaptic coupling using tACS. |
| Dec. 2016 | Speaker, 5th Basic and Clinical Neuroscience Congress, Tehran, Iran
Talk: High-frequency networks determine effective connectivity in neural networks. |

CONFERENCE PRESENTATIONS, POSTERS, SCHOOLS, AND WORKSHOPS

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| Sep. 2025 | Poster, BBI2025 for Non-Invasive Brain Stimulation and Imaging, London, UK
Using SimNIBS Electric-Field Variables to Predict Response to Theta Burst Stimulation Treatment in Males and Females with Treatment-Resistant Depression. |
| June 2025 | Poster, NeuMe Conference, The Royal Ottawa Mental Health Centre, Ottawa, ON, Canada
Theta Burst Stimulation Produces Comparable Clinical Response in Younger and Older Adults with Treatment-Resistant Depression Despite Age-Related Differences in Neuroplasticity. |
| Feb. 2025 | Poster, 6th International Brain Stimulation Conference, Kobe, Japan
Comparing Unilateral and Bilateral Sequential TBS on prefrontal activity: a TMS-EEG study. |
| May 2023 | Poster, 16th Canadian Neuroscience Meeting, Montreal, QC, Canada
Stabilizing post-stimulation oscillatory dynamics by engaging synaptic plasticity with tACS. |
| May 2022 | Poster, 15th Canadian Neuroscience Meeting, Toronto, ON, Canada
Selective engagement of synaptic coupling using tACS. |
| July 2021 | Poster, 30th Annual Computational Neuroscience Meeting, Online |

	Phase response curve of the excitatory-inhibitory neuronal populations and its role in the coherent oscillation of interconnected brain regions.
Dec. 2021	Poster, Neuromatch Conference v4, Online
	Selective engagement of synaptic coupling using tACS.
Oct. 2019	Participant, IBRO-VLTP Courses in Neuroscience, Tarbiat Modares University, Tehran, Iran
June 2018	Poster, Third Workshop on Advanced Methods in Theoretical Neuroscience, Gottingen, Germany
	High frequency neurons determine effective connectivity in neuronal networks.
Jan. 2018	Poster, Computational Neuroscience Conference, IPM, Tehran, Iran
	Manipulating phase relation between neuronal populations by time delay.
July 2016	Poster, 25th Annual Computational Neuroscience Meeting, Jeju, South Korea
	High frequency neuron can facilitate propagation of signal in neural networks.
May 2016	Participant, 3rd Workshop in Computational Neuroscience, Babol, Iran
Dec. 2015	Participant, 4th Basic and Clinical Neuroscience Congress, Tehran, Iran
Aug. 2014	Participant, 4th Computational Neuroscience Workshop on Spikes, Synchronization and Cognition, IASBS, Zanjan, Iran
Oct.-Nov. 2013	Participant, 3rd Tehran IBRO School of Neuroscience: Molecular, Electrophysiological and Behavioral Approaches, Tehran, Iran
Dec. 2012	Participant, 5th Conference in Statistical Physics, Soft Matter and Complex Systems, University of Tehran, Tehran, Iran
Mar. 2012	Poster, Young Atom Opticians Conference, Krakow, Poland
	Single Permanent Magnetic Microtrap.
July 2010	Poster, 22nd International Conference on Atomic Physics, Cairns, Australia
	Single Permanent Magnetic Microtraps for Direct Forced Evaporative Cooling.
Jan. 2009	Participant, Workshop on Quantum Computing and Quantum Information Processing, Shahid Beheshti University, Tehran, Iran
Jan. 2009	Participant, Symposium on Experimental Aspects of Quantum Computing, Shahid Beheshti University, Tehran, Iran

TEACHING AND MENTORING EXPERIENCE

July-Aug. 2022	FUSRP Project Supervisor, University of Ottawa, Ottawa, ON, Canada
July 2020	Teaching Assistant, Neuromatch Academy, Mashhad, Iran
2013-2015	Associate Supervisor, M.Sc. Thesis, University of Zanjan, Zanjan, Iran Student: M. Emamian. Thesis: Honeycomb permanent magnetic lattice for ultra-cold atoms.
Fall 2014-2015	Instructor, Quantum Mechanics, Abdor Rahman Sufi Razi Higher Education Institute, Zanjan, Iran
Fall 2014-2015	Instructor, Basic Physics Laboratory, Institute for Advanced Studies in Basic Sciences, Zanjan, Iran
2012	Workshop Mentor, Numerical Physics for Graduate Students of Physics, IASBS, Zanjan, Iran Software focus: MATLAB.
2012	Instructor, Basic Physics and Laboratory, Industrial Management Institute, Mashhad, Iran
2009-2010	Teaching Assistant, Basic Physics I, University of Zanjan, Zanjan, Iran

TECHNICAL SKILLS

Programming: Python, MATLAB, Fortran.

Software and toolboxes: EEGLAB, Brainstorm, SimNIBS, MATLAB App Designer, NESTApp.

Methods: computational modeling, dynamical systems, neural network modeling, TMS-EEG preprocessing and analysis, EEG/fMRI-informed modeling, effective connectivity, information-theoretic measures, machine learning, and statistical data analysis.

Research methods: non-invasive brain stimulation, tACS/tES/tCS, TMS, theta burst stimulation, neuroimaging-informed stimulation design, and subject-specific modeling.

AWARDS AND HONORS

2018	Q1 Paper Publication Award, Cognitive Science and Technologies Council, Iran
2016	Travel Award for Ph.D. Students, OCNC2016, Jeju, South Korea
